

Adversarial Robustness of U-Net for Inland Water Body Segmentation from SAR Images

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I. Introduction

- ▶ Water-body segmentation from Synthetic Aperture Radar (SAR) imagery supports flood mapping, hydrological monitoring, and environmental assessment. SAR's all-weather, day-night imaging and high spatial resolution make it well suited for complex inland water systems.
- ▶ Although encoder-decoder CNNs such as U-Net [1] perform well, most studies assume clean annotations. In large river basins, masks often contain boundary ambiguities, width variations, and missing segments, introducing label noise. The robustness of segmentation models to such noise remains largely unexplored.
- ▶ Our main contributions include, a framework to simulate annotation errors with morphological operations, a dataset with adversarial examples (available online [2]), and an analysis of U-Net's robustness across corruption levels.

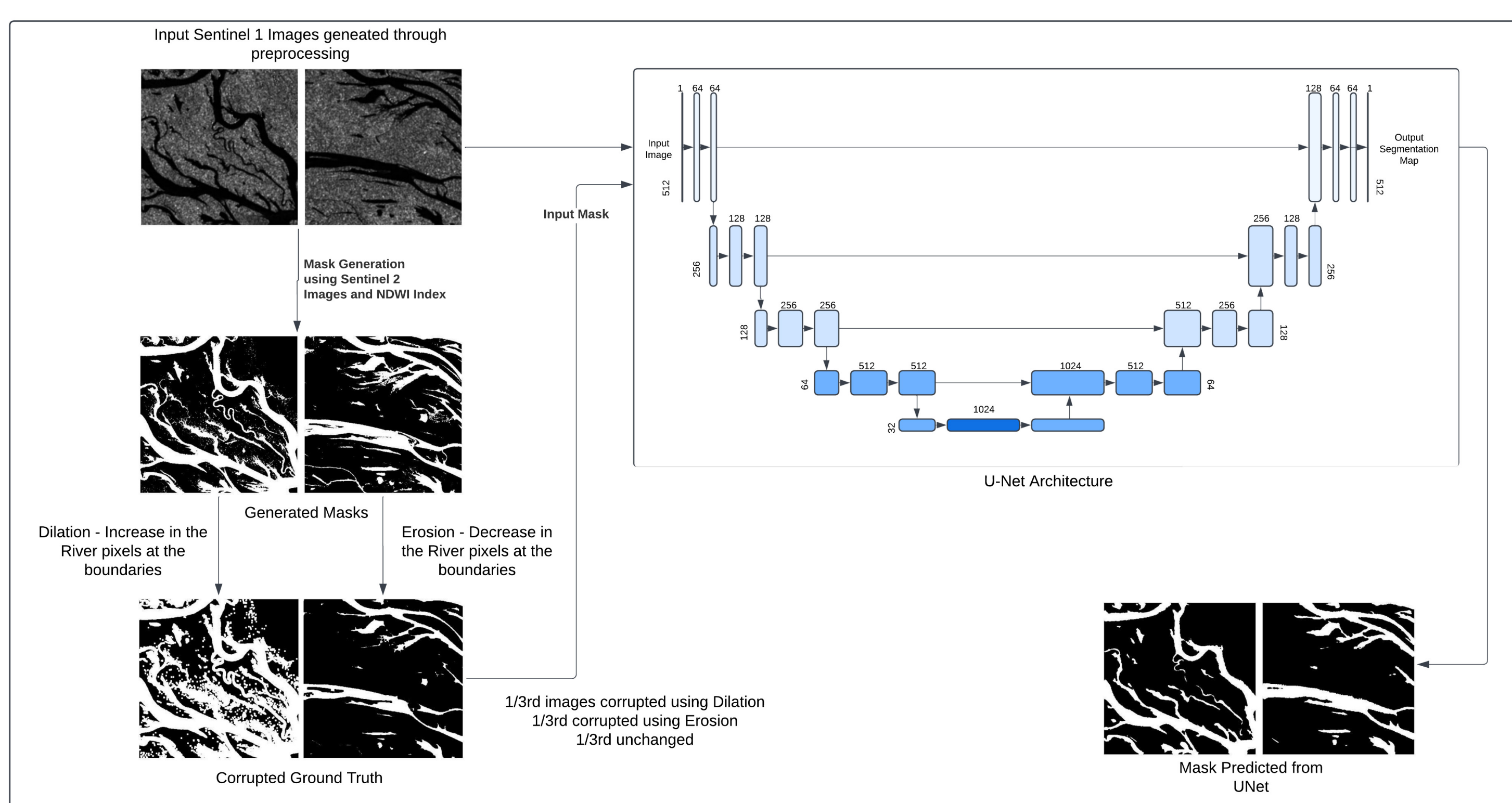


Figure 1: The workflow of our proposed method used for simulating *adversarial attacks* in ground truth generation. We first obtain images and preprocess and clip them, and generate masks using NDWI indices from the corresponding Sentinel-2 images. Morphological Operations are then used to get the corrupted data, which are used as model inputs. The predicted masks post model training are then evaluated.

III. Results

Table 1: Performance of the baseline U-Net model on our dataset at various corruption levels

Corruption Level	Dice Coeff	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
0%	0.770	0.823	0.896	0.869	0.865	0.971	0.955
2%	0.775	0.825	0.896	0.873	0.868	0.968	0.955
12%	0.769	0.822	0.899	0.864	0.865	0.975	0.955
15%	0.768	0.819	0.890	0.866	0.863	0.971	0.954
17%	0.758	0.814	0.903	0.854	0.859	0.979	0.952
20%	0.725	0.790	0.902	0.830	0.841	0.981	0.946
25%	0.597	0.708	0.875	0.759	0.768	0.984	0.920
30%	0.253	0.496	0.818	0.588	0.567	0.996	0.828

Table 1 presents the evaluation metrics of the U-Net model across different corruption levels. The model achieves around 95% segmentation accuracy, with performance largely unaffected up to 20% corruption and only moderate degradation beyond 25%. Figure 2 further illustrates this effect: as erosion increases, the proportion of white pixels decreases, yet the U-Net output, though blurred before thresholding, still captures the river geometry accurately, confirming the model's robustness to substantial corruption. Thresholding restores clear masks until 20% corruption, but beyond that, the model predictions are poor (30%).

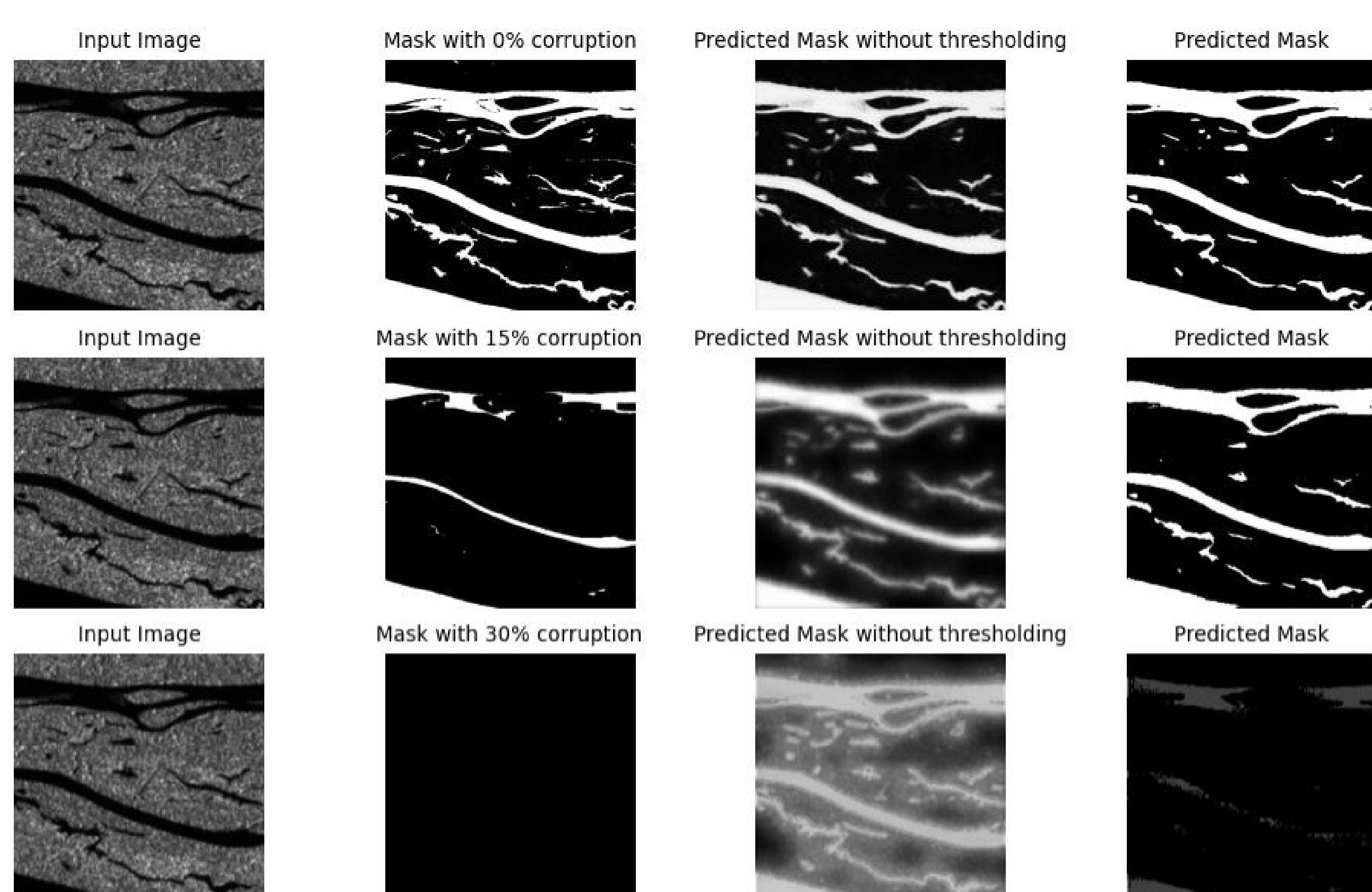


Figure 2: Qualitative results showing the effect of corruption level on the ground truth (second column from the left) and the final predicted mask (rightmost column) mask, for the input image shown (leftmost column). The predicted masks before applying the threshold (second column from the right) to the U-Net model outcomes get blurrier with an increase in corruption, but are robust in capturing river structure in the presence of morphological label noise.

II. Method

- ▶ We generate our dataset using Sentinel-1 SAR imagery for the Amazon Basin, with water masks obtained from NDWI thresholding of corresponding Sentinel-2 MSI data. We chose the Amazon Basin due to the geomorphological diversity shown by the region [3].
- ▶ Example SAR images and masks are shown in the flowchart in Figure 1. The complete data preparation follows standard SAR preprocessing steps [4].
- ▶ To model annotation errors, we apply erosion and dilation to create controlled corruptions. These operations mimic boundary errors common in complex river systems by adding or removing pixels at land-water interfaces.
- ▶ Randomized kernel sizes introduce non-uniformity, and corruption is applied at the levels listed in Table 1. The corrupted masks form the basis of our adversarial experiments.
- ▶ Out of the train and the validation images, we select one-third of the images each for erosion, dilation, and uncorrupted. For an experiment, we then set a threshold on the number of corrupted pixels allowed. Erosion and dilation are applied to the corresponding masks iteratively, till either 100 iterations are over, or the corruption crosses the threshold, following which we choose the previous highest corrupted version.

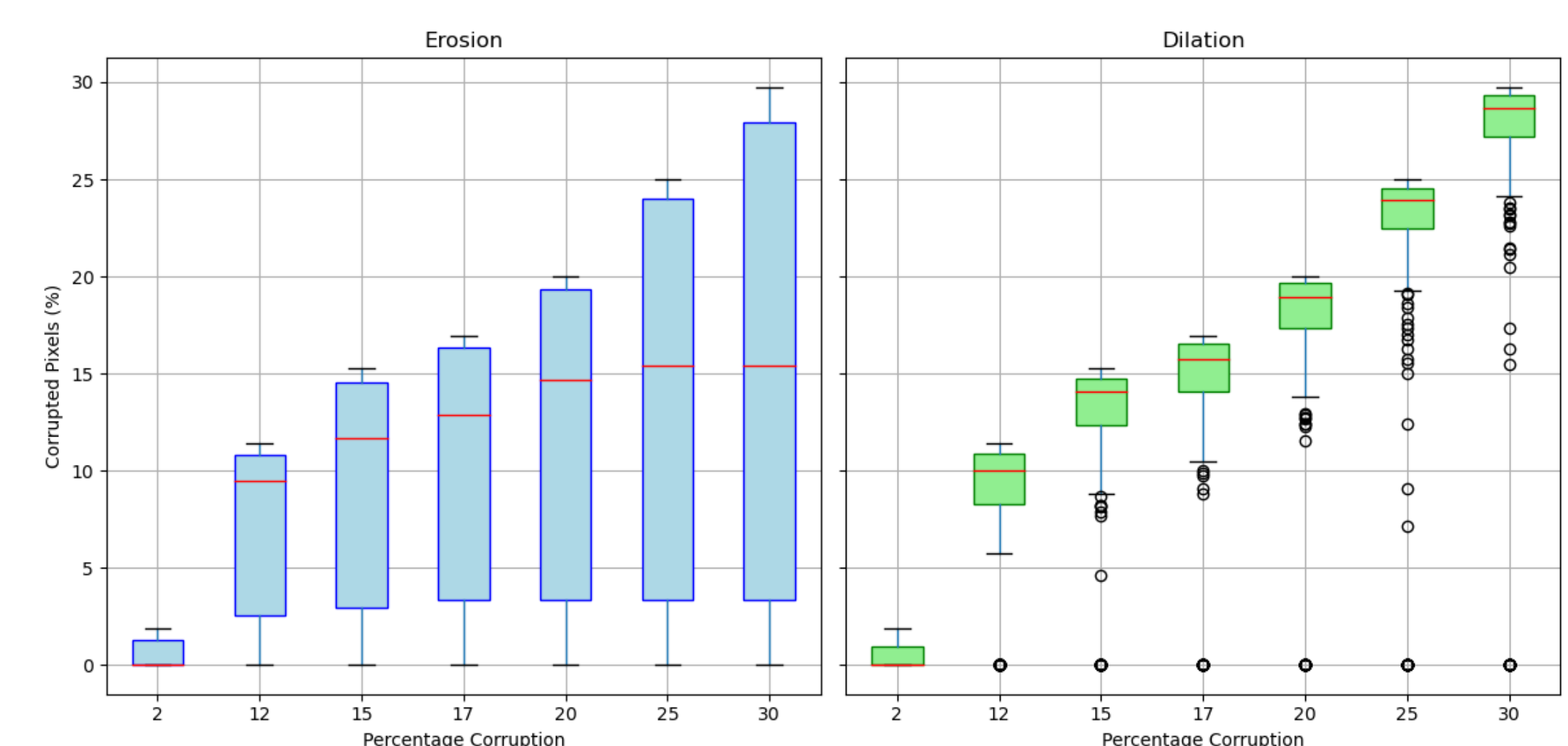


Figure 3: A box plot showing the range of actual corruption in the training data for different corruption levels.

- ▶ Although we set upper limits on corrupted pixels, these are not always achieved, especially for erosion. Since erosion reduces white pixels, it saturates once too few remain to erode further.
- ▶ This behaviour is shown in Figure 3, where box plots illustrate the corrupted-pixel distribution across all training images. Erosion stagnates beyond 20%, with median and lower quartile values changing little, whereas dilation continues to increase as expected.

IV. Conclusion and Future Work

- ▶ The model achieves around 95% segmentation accuracy, with performance largely unaffected up to 20% corruption and only moderate degradation beyond 25%. However, beyond that the model predictions are poor (30%).
- ▶ We selected U-Net for adversarial experimentation due to its strong performance with smaller training sets, faster training, and stability to annotation errors.
- ▶ Currently, more benchmark experiments are being done on the dataset to detect the presence of any anomalous data points.
- ▶ Future work includes extending experiments to other segmentation architectures (including transformer-based architectures), and exploring more traditional corruptions such as rotation and translation.

IV. References

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Proceedings of 18th International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2015), part III 18*, pages 234–241. Springer, 2015.
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- [3] Céline Jézéquel, Thierry Oberdorff, Pablo A. Tedesco, and Laurent Schmitt. Geomorphological diversity of rivers in the amazon basin. *Geomorphology*, 400:108078, 2022.
- [4] Siddharth Kothari, Srinivasan Murali, Sankalp Kothari, Ujjwal Verma, and Jaya Sreevalsan-Nair. Adversarial robustness of deep learning models for inland water body segmentation from sar images, 2025.